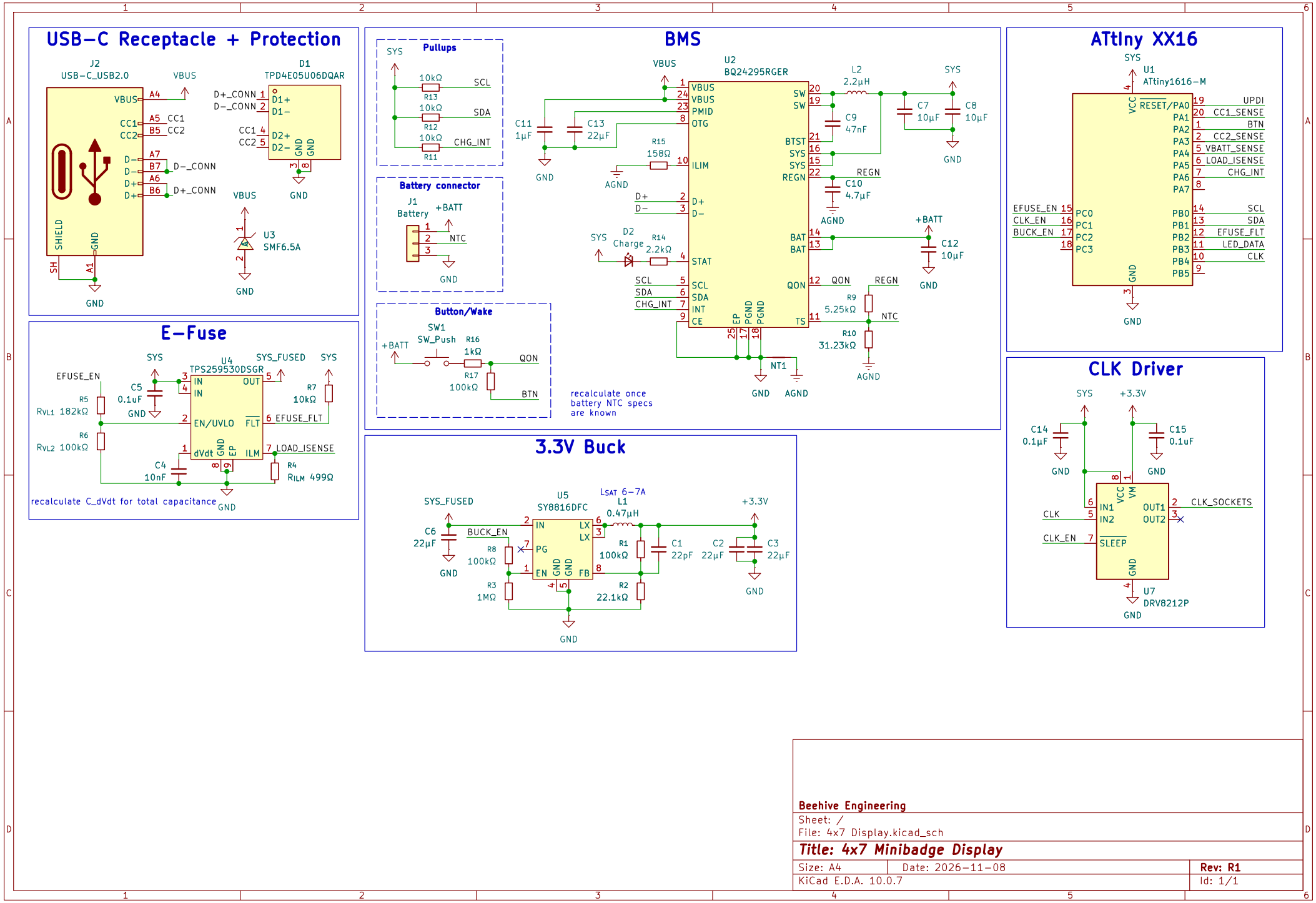

USB-C Receptacle + Protection
J2
USB-C_USB2.0
VBUS
A4
A5 CC1
B5 CC2
A7
B7 D-_CONN
A6
B6 D+_CONN
SHIELD
GND
SH
A1
GND
D1
TPD4E05U06DQAR
D+_CONN 1
D-_CONN 2
CC1 4
CC2 5
D1+
D1-
D2+
D2-
GND
GND
3
8
GND
VBUS
U3
SMF6.5A
1
2
GND
E-Fuse
SYS
U4
TPS259530DSGR
SYS_FUSED
SYS
EFUSE_EN
C5
0.1uF
GND
IN
IN
OUT
R5
RVL1 182kΩ
EN/UVLO
FLT
R7
10kΩ
EFUSE_FLT
R6
RVL2 100kΩ
dVdt
GND
EP
ILIM
LOAD_ISENSE
C4
10nF
R4
RILM 499Ω
GND
recalculate C_dVdt for total capacitance
Pullups
SYS
10kΩ
R13
SCL
10kΩ
R12
SDA
10kΩ
R11
CHG_INT
Battery connector
J1
Battery
+BATT
NTC
GND
Button/Wake
SW1
SW_Push
+BATT
R16
1kΩ
QON
R17
100kΩ
BTN
BMS
VBUS
U2
BQ24295RGER
C11
1μF
C13
22μF
GND
R15
158Ω
AGND
ILIM
D+
D-
SYS
D2
Charge
R14
2.2kΩ
STAT
SCL
SDA
CHG_INT
INT
CE
VBUS
VBUS
PMID
OTG
SW
SW
BTST
SYS
SYS
REGN
L2
2.2μH
C9
47nF
C7
10μF
C8
10μF
SYS
GND
REGN
C10
4.7μF
AGND
+BATT
BAT
BAT
C12
10μF
GND
QON
QON
REGN
R9
5.25kΩ
NTC
TS
R10
31.23kΩ
AGND
EP
PGND
PGND
NT1
GND
AGND
recalculate once
battery NTC specs
are known
3.3V Buck
SYS_FUSED
U5
SY8816DFC
LSAT 6-7A
L1
0.47μH
+3.3V
C6
22μF
GND
BUCK_EN
R8
100kΩ
R3
1MΩ
IN
PG
EN
GND
GND
FB
LX
LX
R1
100kΩ
R2
22.1kΩ
C1
22pF
C2
22μF
C3
22μF
GND
GND
ATtiny XX16
SYS
U1
ATtiny1616-M
VCC
RESET/PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PC0
PC1
PC2
PC3
PB0
PB1
PB2
PB3
PB4
PB5
GND
EFUSE_EN
CLK_EN
BUCK_EN
UPDI
CC1_SENSE
BTN
CC2_SENSE
VBATT_SENSE
LOAD_ISENSE
CHG_INT
SCL
SDA
EFUSE_FLT
LED_DATA
CLK
GND
CLK Driver
SYS
+3.3V
C14
0.1μF
GND
C15
0.1uF
GND
VCC
VM
IN1
IN2
SLEEP
OUT1
OUT2
GND
CLK
CLK_EN
CLK_SOCKETS
U7
DRV8212P
GND
Beehive Engineering
Sheet: /
File: 4x7 Display.kicad_sch
Title: 4x7 Minibadge Display
Size: A4
Date: 2026-11-08
Rev: R1
KiCad E.D.A. 10.0.7
Id: 1/1